

Computer Vision QB

1. Discuss the role of frameworks like OpenCV and deep learning libraries in building Computer Vision applications.
2. Compare & contrast what makes a Computer Vision problem "easy" versus "hard" with one example of each.
3. Explain CV system by specifying the image source, components, and framework, and explain its working.
4. What is mean by computer vision and explain five applications of Computer Vision across different industries.
5. What are different types of image sources used in CV applications? Explain each with example.
6. Explain different geometric transformations (translation, rotation, scaling, reflection, shear) with suitable diagrams.
7. Why does histogram equalization sometimes introduce noise in an image? Analyse the reasons.
8. Explain Counter Tracing algorithm in detail with example.
9. Explain geometric primitives such as points, lines, curves, and shapes.
10. What is the purpose of histogram equalization in image processing?
11. Describe how augmented reality uses camera models, feature detection, and homographies for overlaying virtual objects.
12. What is image warping? Give examples of its applications.
13. Explain image warping and geometric transformations. Provide examples like rectification and morphing.
14. Explain the Hough Transform for line detection.
15. Explain the pinhole camera model. Discuss intrinsic and extrinsic parameters?
16. What are geo-tagged im Histogram equalization in image processing ages? How are they used in CV applications?
17. Explain the concept of Bag-of-Words (BoW) model for image representation
18. What is 3D reconstruction in computer vision? What are some real-world applications of 3D scene reconstruction?
19. What is Structure from Motion (SfM)? Explain its main objective.
20. Explain Convolutional Neural Network? Explain the steps involved in training a CNN model.
21. Explain challenges faced in implementing SfM in real-world scenes.
22. Explain Structure From Motion
23. Explain the architecture of a Convolutional Neural Network (CNN). Describe the role of convolution, pooling, and fully connected layers.
24. Explain Eigenfaces – Definition and Use in Face Detection
25. Explain Convolutional Neural Network? Explain the steps involved in training a CNN model.

26. Explain the Viola-Jones algorithm is used for face detection? What are the components of this algorithm
27. Explain Haar-like features in face detection and how they can detect different facial characteristics?
28. Explain how AdaBoost algorithm works in face detection.
29. Compare face detection and face recognition. Explain the different challenges associated with both processes.
30. Explain the Advantages of CNN
31. Draw and explain the general architecture of a CNN
32. Describe the steps involved in pedestrian detection. Explain any two feature extraction techniques used for pedestrian detection
33. Explain CNN Layers and it's application